

Challenge Questions

1. Consider the curve $x(t) = e^t$, $y = t + 3$. Eliminate the parameter to find a Cartesian equation of the curve. Then sketch the curve and indicate the orientation of the curve.
2. Let C be the portion of curve of $y = \sqrt{x}$ from $(1,1)$ to $(4,2)$.
 - a) Create two different parameterizations for C .
 - b) Create a parameterization for $-C$ (this is the same as the curve C but traced in the opposite direction).

3. Consider the curve $x = e^t \cos(t)$, $y = e^t \sin(t)$.

a) Make a sketch of the curve for $t \geq 0$ by plotting points. Don't use your calculator.

b) Determine when the tangent line to this equation is horizontal and when it is vertical.

c) Find the arc length for this path for t from 0 to 1.

4. Consider the curve $x(t) = 2t^2$, $y(t) = t^3 - 4t$.

a) Determine at what value(s) of t the curve reaches the point $(8, 0)$.

b) Find all values of t for which the tangent line is horizontal or is vertical.

c) Use $x'(t)$ and $y'(t)$ together with intercepts and behavior as $t \rightarrow \pm\infty$ to sketch the curve. Don't use your calculator.

d) Find the slope of the two tangent lines at $(8, 0)$.

5. Consider the curve $x = a \cos(t)$, $y = b \sin(t)$.

a) Show that this curve models a particle traveling counterclockwise about an ellipse starting at the point $(a, 0)$. Recall that the general form for the equation of an ellipse centered at the origin is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

b) Determine for what t the tangent line has slope $\frac{a}{b}$.

c) Setup, but do not evaluate, the integral for the circumference of the ellipse using this set of parametric equations. This integral is an “elliptical” integral and is another integral that does not have a nice anti-derivative.

6. The parameterization $x = \cos(t)$, $y = \sin(t)$ for $0 \leq t \leq 2\pi$ creates a unit circle with counterclockwise orientation.
- a) Create a parameterization for a unit circle with clockwise orientation.
- b) Create a parameterization for a circle of radius r .
- c) Create a parameterization for the following ellipse.

